

ELEMENT C

In developing an effective solution to prevent water from leaking out of school A/C units and creating hazards on sidewalks, several high-priority design requirements emerged from both research and stakeholder feedback. The first requirement is that the solution must significantly reduce the amount of water leaking from the A/C by redirecting all condensation into a controlled drainage system. It must be easy to attach and remove without damaging the unit, durable enough to last eight to twelve weeks, and equipped with both a drainage tube and an overflow hole to prevent water buildup. This requirement is grounded in research showing that many previous drainage attempts, such as under-sidewalk drains and French drains, failed because they clogged or overflowed. When I shared this idea with a custodial staff member, they emphasized the need for easy removal and warned that added weight could damage the A/C. Their feedback led me to revise the requirement to include removability and a drainage tube to prevent water from becoming too heavy. A second high-priority requirement is that the solution must be able to handle at least 2.5 gallons of water per day. Research shows that window A/C units can produce more than two gallons of condensation daily, especially during humid seasons. Because fall humidity increases water output, the system must be capable of handling more than the minimum amount. After sharing this requirement with [REDACTED] he recommended increasing the capacity to account for seasonal humidity changes. Based on his feedback, I revised the requirement from two gallons to 2.5 gallons per day to ensure the solution remains reliable throughout the year.

The third high-priority requirement is that the solution must be made from water-resistant or fully waterproof materials that can withstand continuous exposure to moisture for at least two months. This requirement ties directly to research showing that many drainage systems fail because materials weaken, rust, or mold when exposed to constant moisture. Since the A/C will release water daily, the solution must resist corrosion, mold growth, and structural weakening. When I shared this requirement with a maintenance staff member, they confirmed that water-resistant materials are essential and noted that plastic or rubber materials work best because metal rusts. Their feedback supported the requirement, so no revisions were needed. Another high-priority requirement is that the solution must be installable by one person in under ten minutes using only basic tools such as a screwdriver or zip ties. This requirement is supported by research showing that many prior design attempts like French drains, rain gardens, and sloped sidewalks required heavy labor, professional installation, or major construction. School maintenance staff need solutions that are quick, simple, and do not interrupt school operations. When I shared this requirement with custodial staff, they explained that installation must be fast because they do not have time for complicated setups. Based on their feedback, I added the measurable requirement of a ten-minute installation time and limited the tools needed to basic, commonly available ones.

The fifth high-priority requirement is that the A/C system must use a sealed, correctly sized drain line at least twelve inches long that can handle the unit's flow rate and redirect all water to a safe drainage point so the sidewalk never becomes wet. This requirement connects to research showing that clogged or poorly placed drain lines are the main cause of leaks. Prior designs like under-sidewalk drains failed because water still reached walking surfaces, creating slipping hazards. A sealed, properly sized drain line prevents this by ensuring water is fully contained and directed away from high-traffic areas. When I shared this requirement with a school administrator, they emphasized that the sidewalk must stay dry at all times for safety. Their feedback confirmed the importance of this requirement, so no revisions were needed. Two lower-priority requirements also emerged from the research and stakeholder conversations. The first is that the solution should include an accessible inspection point such as a clean-out port or viewing window so maintenance staff can quickly check for blockages or leaks without removing panels or disconnecting equipment.

This requirement ties to research showing that clogging was a major failure in prior designs like French drains and under-sidewalk systems. An inspection point allows quick checks and prevents unnoticed buildup that could lead to leaks. Maintenance staff agreed that a quick visual check would save time, so no revisions were necessary. The final requirement is that the solution should include a secondary drainage line at least twelve inches long and a backup catch pan with a minimum depth of two inches to redirect water safely if the main drain becomes blocked. This requirement connects to research showing that many drainage systems fail when a single drainage path becomes clogged. A backup system ensures the sidewalk stays dry even if the main line fails. Custodial staff agreed that backups are helpful because clogs happen more often than people realize, confirming the need for this requirement. Together, these design requirements create a clear, research-based, and stakeholder-validated foundation for developing a safe, durable, and effective solution to the A/C water leakage problem.